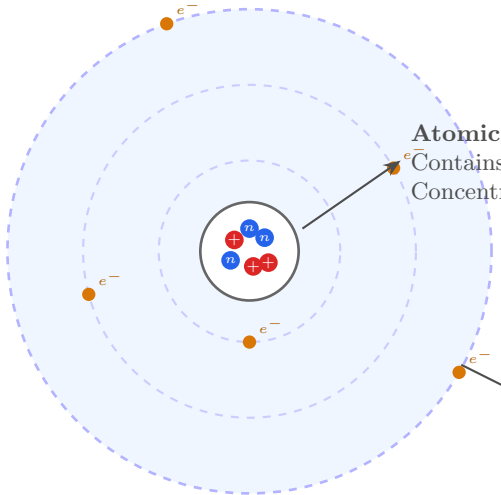


Atomic Nucleus ($r \sim 10^{-14}$ m)
Contains Z protons (p^+) and N neutrons (n^0)
Concentrates $> 99.95\%$ of atomic mass

Electron Cloud ($r \sim 10^{-10}$ m $= 1$ Å)
Negligible mass ($m_e \approx m_p/1836$)
Defines chemical volume and reactivity

Scale Comparison: The atom is $10\,000\times$ larger than its nucleus ($\approx$ a marble at the center of a cathedral!).